Enhancing Pharmacy Inventory management systems in Rwanda for Real-Time Patient Access to Medicines .

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July 2026

Problem Statement

Patients in Rwanda face significant economic and time burdens when searching for essential medicines due to frequent stockouts (Bizimana et al., 2020). There is a high probability that a patient will not find their medicine because the availability rate in Rwandan pharmacies is far below the World Health Organization's recommended 80% standard (Biziman et al.2020),This means that nearly 1 in every 3 pharmacy visits ends in failure, forcing patients to waste time and money searching elsewhere..

The consequences of this information gap are substantial. Patients must physically visit multiple pharmacies, a process that consumes valuable time, incurs transportation costs, and delays treatment initiation. This creates a significant hidden economic burden for patients, who must pay for repeated transportation as they search for their medicine.

While Rwanda has implemented inventory management systems like eLMIS to help public health facilities to track how many pills, antibiotics are available in clinics across the country (Ministry of Health, Rwanda, 2018) and private pharmacies utilize software such as Ishyiga for stock management and inventory

tracker (WazoPlus, 2024), these systems remain siloed and inaccessible to patients. This creates a critical market failure where real-time stock data exists within pharmacy systems but cannot be utilized by those who need it most—patients seeking medications.

Globally, some African countries , such as Cameroon and Ethiopia , have already developed digital solutions to address the issue of finding medicines. In Cameroon , a platform called Urpharm which is currently used has a mission of facilitating patient access to medication, allowing them to easily locate and purchase them with just a few clicks (Urpharm, 2020). Similarly, in Ethiopia, the Medhanite app-based pharmacy service was launched in July 2022 to help users find medications in Addis Ababa(Solomon, 2022). These systems already enable patients to search for nearby pharmacies with specific medicines, and this reduces the effort, time and money patients spend searching for medicines in the capital(Solomon, 2022).

However, no integrated, patient-facing platform currently exists in Rwanda to address this gap . This research therefore addresses a dual challenge: first, to quantify the patient-level burden of medicine stockouts in terms of time and transportation costs, and second, to develop and evaluate a digital solution that

bridges this critical information gap in Rwanda's pharmaceutical sector. This will contribute to Rwanda’s broader healthcare objectives, as outlined in the National Digital Health Strategic Plan(Ministry of Health, 2018).

Proposed Solution

As a solution, this project aims to design, develop, and pilot a digital pharmacy inventory management platform that enables patients to locate nearby pharmacies that have the medicines they need in stock, while allowing pharmacies to efficiently update and manage their inventory data in real time. The system is intended to improve access to essential medicines by bridging the information gap between pharmacies and patients in Kigali, Rwanda.

The pilot implementation will take place in Kigali, Rwanda, focusing primarily on Nyarugenge District, specifically around the City Tower area. This location was selected because it hosts a high concentration of pharmacies and diverse patient needs, making it a suitable location for real-world testing the system’s functionality and effectiveness.

The user scope for the pilot will be conducted in Nyakabanda sector(Nyarugenge District) near the place I live , as well as in other areas within Kigali in town where many people are concentrated. This ensures that the pilot will work well both in

the local urban community and other busy parts of Kigali, creating more realistic and diverse testing environments.

The development of the platform's core features is expected to be completed in approximately 1.5 months before the pilot phase begins .During the pilot, 10 to 20 patients will use the platform to search for medicines . Around 4 pharmacies near City Tower will participate by updating their inventory either manually or through CSV uploads.

The core functionality of the platform during this pilot includes real-time inventory updates, patient search for medicines, and the display of nearby pharmacies on a map or list. There is potential for future integration with existing pharmacy management systems, such as Ishyiga POS .

The pilot testing will last for three months, during which continuous feedback will be collected from both patients and pharmacy staff . This feedback will be analyzed to assess system usability, accuracy of displayed inventory, and overall improvement in access to medicines

Future enhancements may include integration with existing pharmacy management systems such as Ishyiga POS, allowing for automatic synchronization of inventory data across multiple pharmacies. The pilot phase will serve as a proof

of concept, ensuring that the system is technically feasible, user-centered, and scalable for potential expansion across Rwanda.

Key Features & Target Users

1. Real-Time Medicine Search & Discovery
2. Feature: An intuitive search bar allowing patients to find pharmacies with the medicine by name, with instant results.
3. Implementation: Built with React & TypeScript for a fast, responsive interface. Connects to a Django REST API that queries the database in real-time.
4. Impact: Patients can find available stock in seconds, eliminating the need for physical searches.

Technology Stack

Frontend Development

- React with Vite & TypeScript
- Tailwind CSS

Backend &API development

- Django & Django REST Framework

Database

- SQLite

Design & Prototyping

Figma

Testing & Quality Assurance

- Vitest: For comprehensive unit testing of core functions and API logic.
- Chrome DevTools: For performance and debugging.
- Cypress: For end-to-end testing of user workflows and integration